

Body as Starting Point 5: Exploring the Inbodied Interaction Design Framework - New Methodologies in Interactive Health Design

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ABSTRACT

We invite you to celebrate the fifth inbodied interaction workshop at CHI by exploring the Inbodied Interaction Framework to align your designs with the internal complexity of the human body's interconnected, physical, and biological networks first with the goal to "#makeNormalBetter" for all at scale. This year we are introducing the new Inbodied Interaction Design Framework with a set of guiding questions and provocations to lean on and re-invent working practices. In this virtual workshop, we welcome participants with no prior experience with inbodied interaction and familiar practitioners who want to gain an alternative perspective for technology design that takes the body as a starting point.

CCS CONCEPTS

• **Human-Centered Computing - Human Computer Interaction (HCI);**

KEYWORDS

Inbodied Interaction, Wellbeing, Human Performance, Health Design

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1 BACKGROUND

At CHI 2022, we are celebrating the Fifth *Body as Starting Point Workshop on Inbodied Interaction* at CHI. Over the past five years, we have been exploring what happens to our interactive technology

designs when we align our designs with the internal complexity of the human body's interconnected, physical, and biological networks first with the goal to "#makeNormalBetter" for all at scale.

Inbodied Interaction (II) is about how to align our designs of computational, interactive systems with how we work, as complex physical, electrical, chemical, neurological (ie – inbodied) systems. As an Approach – II, pronounced *Aye Aye*, postulates that if we align our designs with that inbodied wiring, then we create better designs – that support our performance – including health and wellbeing – with less effort. Inbodied Interaction also foregrounds that the body is the site of constant adaptation. These adaptations can be positive and healthful or negative and unhealthful or, as we frame it in II, they can be more performance enabling or performance disabling. So something like sitting eight hours a day is well established now to induce more disabling adaptations. From an Inbodied Interaction perspective, sedentarism is the antithesis of aligning the design with how we work, with how our inbodied systems perform best. It creates a very performance-disabling set of adaptations. When we align our designs with how we function best as complex electro-chemical, neuro-social organisms, we build and maintain our health. Inbodied Interaction postulates a set of components to help designers access the in-bodied - our complex neuro physio electro chemical selves - in order to create designs that are more likely to have lasting benefit for those using them.

Our past workshops have helped evolve key concepts for Inbodied Interaction like Tuning, Discomfort Design, in5/c4, Design Continua, and Exits (overview). These explorations, supported by a growing community, has produced a CHI sponsored Summer School on Inbodied Interaction, an ACM Interactions Special Issue [5,11], several CHI Courses on Inbodied Interaction [6, 7], and workshops (e.g. [1, 2, 4, 8, 10, 12]) informing a forthcoming special issue in *Frontiers in Computer Science* (overview).

This year, we are bringing together inbodied interaction themes to a new InbodiedInteractionDesignFramework. While we have developed inbodied concepts for design, this is the first time we have proposed a formalism of how to craft a design space in which to use those concepts. Our goal in the workshop is to test out the Design Framework, across design scenarios and from these tests, to improve it towards making it as easy as possible for designers to *align our designs* with our internal selves - our inbodied selves.

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The workshop welcomes participants of all experience levels, from no prior engagement with inbodied interaction to familiar practitioners. To apply, we invite you to engage with one of the presented scenarios in this paper and explore the Inbodied Interaction Design Framework and submit your one-page exploration.

1.1 Outcomes Include

- Increase the number of people engaged with inbodied design in their work. This will be measured by the number of people who join our thriving slack community. We aim for 20 new members.
- Revise our II design framework based on input from people new to II design, and people experience with II design. Novices will help us understand how to explain concepts. Experienced people will help us revise concepts. This will be measured by tracking the number of changes made to the framework. We aim to revise about one fourth of the framework.
- Increase the visibility of II in the broader research community. This will be measured over time by the number of papers involving II that appear in print. We aim for four papers in print in the 18 months following the workshop.
- Increase understanding of II in design. The concepts in II are meant to be understood by people with no specialized training in physiology and related fields but must be understood to be used correctly. Progress on this goal is difficult to measure and we will not measure it, but it is perhaps the most important goal of the workshop.

2 ORGANISERS

Josh Andres is a Senior Lecturer in the School of Cybernetics at the Australian National University (ANU). He is one of the founding members of the inbodied interaction movement, having co-lead the previous four CHI workshops, the summer school, and now as one of the editors in the upcoming Frontiers special issue on inbodied interaction. Josh's research advances human-cyber-physical AI systems using a cybernetic lens and drawing from human-computer integration and inbodied interaction paradigms. Josh is part of the ACM SIGCHI Executive Committee as Adjunct Chair for Awards, an associate with the WellthLab at Southampton U. UK, and the Exertion Games Lab Monash U. AU. His Twitter handle is `<ext-link xmlns:xlink="http://www.w3.org/1999/xlink" ext-link-type="twitter" xlink:href="https://twitter.com/ExperiencePlay">@Experienceplay</ext-link>`

Richard Gomer is a lecturer in Computer Science at the University of Southampton, UK. His research interests centre around how technology that extends and changes human agency, including within domains such as health and wellbeing.

Michael Jones Professor of Computer Science at Brigham Young University. His research interests include the use of sensor data in sports training, interactive systems in outdoor recreation and discomfort as a design material. He has co-organized workshops at CHI on HCI outdoors and Inbodied Interaction. He co-edited a book on HCI outdoors based on the CHI workshop.

m.c. schraefel holds the position professor of computer science and human performance at the University of Southampton, UK,

and director of the wellthlab.ac.uk, where the vision is to help #makeNormalBetter for all, @scale. The mission is to explore how interactive technologies can be designed to support that mission – and how using an inbodied interactive approach in particular can help inform those explorations.

3 WEBSITE

The Workshop Website will be at <https://wellthlab.ac.uk/ii5>. It will provide guidance for the pre, during and post workshop activities, detailed below.

4 EXPLORING NOVEL INBODIED INTERACTION DESIGN FRAMEWORK

Each year we have had a specific workshop focus around developing an area of inbodied interaction. This year our focus is on how the Inbodied Interaction Design Framework can be applied to design challenges. This will allow us to test the framework itself, to refine it, and to create examples of application. This will be valuable not just for the inbodied interaction community but also for the broader health interaction design communities.

A key design opportunity we have found in Inbodied Interaction is to:

- (1) align aspiration with the intervention. And then,
- (2) align the design with our inbodied selves - leveraging the previously developed inbodied principles of in5/c4.

For point 1, *align aspiration*, we begin a design process by considering the aspiration without a predetermined intervention. For example, rather than suggest to someone that they need to stand more at work (an intervention as starting point) – which could be part of a strategy for health since that aligns with our physiological need to move for various internal processes, it may not be an optimal strategy for greatest benefit for all affected. We suggest starting with an aspiration rather than practice. If a workplace is where the intervention is to take place, we would explore: how do people aspire to be when at work? Once we understand those aspirations, we can consider appropriate interventions.

For point 2, *align design with inbodied*, we offer a related critical question: given the aspirations of say this work community to be healthier, or smarter or happier and individuals to be less tired, be stronger, and be happier, how can we support these values within a workplace context drawing on inbodied concepts from in5, to discomfort to tuning? If standing is to be part of a strategy, are their cultural barriers to that? Where might technology assist making connections between the physical benefits of in5 movements, including standing, and feeling more capable at work?

4.1 The Skeleton Check List of the Design Framework

For the workshop, we will be asking participants to test the SkeletonDesignFrameworkavailableonlinehere. This template includes the following inbodied interaction design series of questions:

What is the adaptation you seek

This adaptation can be anything: to be calmer; becoming more creative; learning a new language; getting faster in a 5k run. In the example below, we focused on supporting enhancing creativity.

What is the value around which this adaptation is aligned?

Another way of asking this is to unpack what are the attributes within the sought adaptation? e.g., in our creativity example below, better creativity for the folks working at an advertising agency could translate into better campaigns, promotion, opportunities, success.

How do you evaluate success?

There may be personal markers of success like interrupting less in conversations; running faster; meditating longer. In many cases there are established assessments that can be used over the course of an exploration. We can also consider both qualitative and quantitative assessments as well as the frequency of these assessments. There may also be success criteria that are important for you as a designer rather than just benefit for participants. Such as: for insourcing with a time-limited intervention – are the participants still using the knowledge, skills and practices from the study weeks, months, years later?

What is the imagined life cycle of this interaction?

Within this question is also the EXITSTRATEGYquestion – what is the ending of this technology? In HCI we rarely have time to ask this question. We want to know that it works now for whatever the “now” context is. In inbodied interaction, where we are also interested in quality of life, over the lifecourse, considering the effects of interactions both now, and ten years from now, may be novel – because we are driven by component parts. But – what happens to our designs when we consider interaction longevity as a design affordance/constraint. Indeed, TIME for each phase within the lifecycle – from a macro, meso and micro perspective can also be mapped out explicitly in this method. Here is also where we may consider the EXIT STRATEGY – where the off ramp is – for our design.

Where is the design situated within the CONTINUA?

Where and when in the design space are thecoreIIcontinua (See [9, 13]) of
insource-outsourced
heuristic-habit
raw-cyborg
feature as deliberate attributes of the design space you’re building for your intervention?

How support TUNING

In Inbodied Interaction, tuning considers three components: (1) an experience (2) how that experience is mediated by Knowledge Skills and Practice (KSP) and (3) how it comes to be felt internally (consciously perceived and interpreted). Tuning is about exploring how to dial in each of these components,

across contexts, to support optimal adaptation via connecting KSP and perception, towards enabling performance. Thus, questions to be explored, on the FELT side of the Tuning experience: what are the signals and perceptions around which we want to develop awareness? What are the physiological processes we can draw on to support this?

Related Questions to be explored, on the ADAPTATION side of the Tuning experience: What are the Knowledge Skills and Practice (KSP) parts to help support, dial in, refine that awareness? and improve the experience? in other words how is KSP connected to inbodied awareness of the experience the KSP is helping to create? HOW do we design interventions to build these connections?

Where are the in5 and c4 in this design?

Each of theinbodied5(in5) andcircumbodied4(c4) are interconnected. As all experiences are filtered via the current state of the inbodied – of our homeostatic state – then the in5 become the foundation of any other adaptation or practice. They also complement adaptations and processes. We may in a design focus on one particular attribute, but where the larger design map may consider at first where each of the in5/c4 have a role in whatever the adaptation may be. We can walk through EACH of the in5, c4, to consider the two sides of the TUNING experience – awareness of the internal felt experience of these parameters and how improving KSP of these parameters may better support the adaptation desired. By doing such a walk-through, we can explore which attributes are of particular relevance for our intervention.

What are the Discomfortable Parts?

A key part of II is the role of discomfort as supporting positive adaptations. it’s useful to dig into what makes the practice – whether insourced or outsourced – difficult to the point of being unappealing, repulsive, averted? In physical exercise, we often hear “well if that doesn’t appeal to you find something that does; the main thing is you stick with it”. So, people just go run or just go lift weights. Great. but for overall health and longevity, a preference-driven practice is likely insufficient. Escaping discomfort is not life. How can we prepare for that discomfort and use that as a design material?

5 WORKSHOP PLANS: VIRTUAL BY DESIGN

Given the uncertainty of travel this year, we are committed to a maximally inclusive, virtual workshop. Last year our virtual approach had a strong turnout and great engagement. We will build on that success for this anniversary meeting.

5.1 Pre workshop Plans

We offer a collection of provocative questions to re-invent working practices and invite participants to lean on the InbodiedInteractionFramework to explore them.

5.2 The UnOffice for the Future of Work: Contact, Vision, Balance – the Parkour Office?

There is growing acceptance about the problems of sedentarism at work – that sitting all day is causal – not just correlated – with a host of negative side effects. What is less often discussed are the host of designs that are unaligned with human interaction.

Question one: How might we incorporate new ‘normal’ office practices aligned with the inbodied to promote creativity and trust-building?

For example, through various COVID lockdowns, people in conversation and in the media made clear how much they missed being “in person” with others. From an Inbodied interaction perspective, this is predictable. Physiologically (as per the in5 ENGAGE) we are wired to be physically in proximity with others [14]. We have multiple dedicated nerves and fine muscles to support facial expressions – the roles of which are to support communication. And yet the benefits of reducing transportation costs of virtual meetings are indisputable.

Question two: How might we better amplify physiological signals to be perceivable in virtual context?

A negative side effect of more close focus screen time, however, is that it keeps us in that fixed sedentary position, with most of our movements in the sagittal plane – even if standing. Close focus screen work is associated with increasing global levels of myopia [3], and yet our eyesight is the highest order perceptual system we use to navigate and sense make the world. What if we were to reject the cost we pay with our eyes of screen/text technology?

Question three: How might we reinvent “reading” in particular to leverage our “relaxed” focal position that is about at 7 feet from us?

Most workplaces that are indoors are particularly sterile when it comes to microbiome diversity (c4-microbiom). Even walking meetings on a large campus can be denatured, with unnatural light cycles taking us out of diurnal rhythms.

Question four: How might interactive technology help us re-imagine work itself so that getting outside may not be a break but a contribution to work?

5.3 How to Join the workshop

We invite participants to explore alone or in groups one or as many questions as they wish using the InbodiedInteraction\penalty-\@MDesignFramework. Submissions should include a description, illustration, or video of the artefact designed to address their situation and walk through how they used the design framework questions to formulate the affordances and constraints of their designs. To submit, visit the website for details <https://wellthlab.ac.uk/ii5>

People interested in participating, prior to their submission, will also be able to join our inbodied interaction slack community. Please email any of the authors your request.

6 WORKSHOP STRUCTURE

6.1 Workshop Structure

Our workshop will take place through a combination of asynchronous activity and the day-long virtual workshop itself. The activities will guide participants through a design process that uses

the inbodied interaction framework to tackle one of our identified challenge areas. Our aims are to educate our participants the II design framework, to expand the II design framework with their insights and experience, and to understand how we can improve our explanations and design tools to make them more accessible to design practitioners.

Our indicative timeline for the day of the workshop is as follows:

10:00 - 11:15 An introduction to Inbodied Interaction Design Framework (IIDF) and the challenge areas

An initial welcome, and presentation about the goals of the IIDF and the process for the day. Followed by groups working together to understand a challenge area using the IIDF as a lens, and empathising with potential target users. Followed by group ideation about inbodied intervention strategies that are aligned with the in5 and c4.

11:15 - 11:30 BREAK

11:30 - 13:00 Continua, phases and exit strategies

A second presentation will introduce more concepts from the IIDF: continua, phases and exit strategies.

Participants will develop ideas from the first session to situate them within the II continua, identify phases within the emerging interventions and plan possible exit strategies.

13:00 - 14:00 BREAK

14:00 - 15:30 KSP and discomfort design

The third session will focus on introducing the Knowledge Skills and Practice (KSP) element of the IIDF, along with discomfort design. Participants will use these concepts to develop their design concepts further; annotating the intervention timeline with how KSP and discomfort might be used.

15:30 - 15:45 BREAK

16:00 - 17:00 Reflections on the IIDF

We will conclude the workshop by reflecting on how the IIDF was used by participants, and where it could be developed further. We'll encourage participants to think about how it could be used in their own work, and identify possible collaborations to develop or apply elements further.

6.2 Post Workshop Plans

Our main goal from the workshop itself is to generate worked examples of how the inbodied interaction framework can be used and improved to make it as direct as possible to create and evaluated designs that are aligned with how we work as inbodied creatures. Post workshop, these outputs will be made available on our community site. These will provide a community resource of shared, worked examples for both research and teaching. We will also continue conversations in slack to support the community's exploration and development of inbodied interaction based on these outcomes. And, as demonstrated with our track record on translating workshop outcomes into future publications and events that disseminate the learnings, we plan to do the same with the outcomes from this workshop.

7 CALL FOR PARTICIPATION

We invite you to celebrate the fifth inbodied interaction workshop at CHI by exploring the Inbodied Interaction Framework to align your designs with the internal complexity of the human body's

interconnected, physical, and biological networks first with the goal to "#makeNormalBetter" for all at scale.

All experience levels welcome

We invite participants to explore alone or in groups one or as many of the provocations presented in this paper using the InbodiedInteractionDesignFramework. Submissions should include a description, illustration, or video of the artefact designed for the provocation and a walkthrough of how they used the framework to formulate the affordances and constraints of their designs. To submit, visit the website <https://wellthlab.ac.uk/ii5>

At the workshop, virtual design jams will take participants through a rapid design process using the inbodied interaction design framework to design conceptual interventions for the provocations. The outcomes will be a new resource for the health interaction design community to see how this framework can help design and validate health-supporting interactive technologies.

All the previous CHI workshops sold out, and we hope to keep the group small enough for an intimate experience; please apply early to confirm your place.

Inbodied interaction is an emerging area in HCI that offers participants the opportunity to break new ground, with the support of a growing community of HCI and adjacent domain experts, to design interactive interventions that take the body as a starting point to facilitate human performance aspirations.

Thank you, from: Josh, Richard, Mike, and m.c.

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